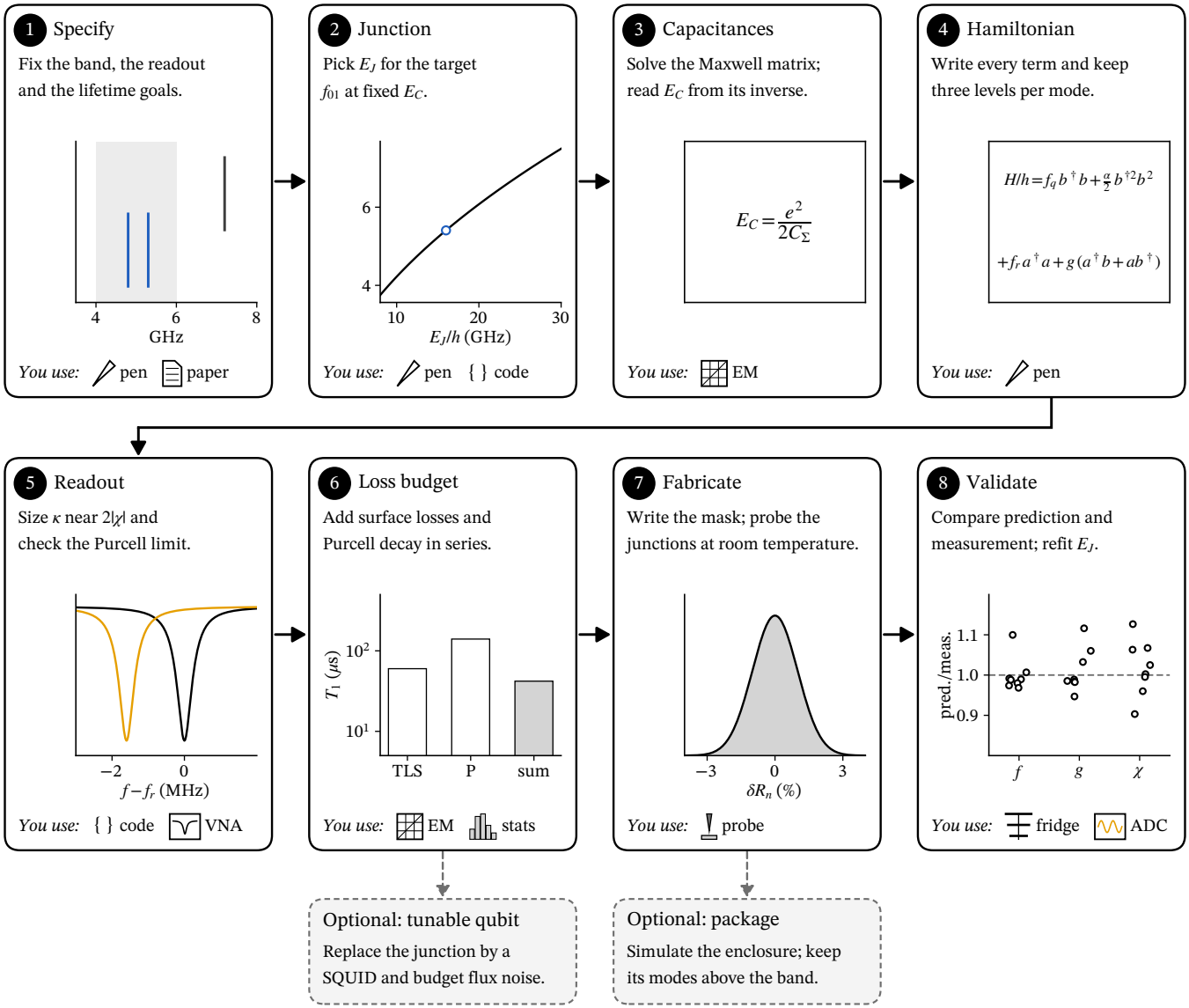
1 Specify
Fix the band, the readout and the lifetime goals.
4
6
8
GHz
You use: pen paper
2 Junction
Pick E_J for the target f_{01} at fixed E_C.
6
4
10
20
30
E_J/h (GHz)
You use: pen { } code
3 Capacitances
Solve the Maxwell matrix; read E_C from its inverse.
$E_C = \frac{e^2}{2C_\Sigma}$
You use: EM
4 Hamiltonian
Write every term and keep three levels per mode.
$H/h = f_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2$
$+ f_r a^\dagger a + g(a^\dagger b + a b^\dagger)$
You use: pen
5 Readout
Size κ near $2|\chi|$ and check the Purcell limit.
−2
0
$f - f_r$ (MHz)
You use: { } code VNA
6 Loss budget
Add surface losses and Purcell decay in series.
T_1 (μs)
10^2
10^1
TLS
P
sum
You use: EM stats
7 Fabricate
Write the mask; probe the junctions at room temperature.
−3
0
3
δR_n (%)
You use: probe
8 Validate
Compare prediction and measurement; refit E_J.
pred./meas.
1.1
1.0
0.9
f
g
χ
You use: fridge ADC
Optional: tunable qubit
Replace the junction by a SQUID and budget flux noise.
Optional: package
Simulate the enclosure; keep its modes above the band.